Date	03 July 2026
Team ID	SWTID–2026–4593
Team Size	3
Team Leader	Hari Bellamkonda
Team Member 1	Kesha Vardhan Chinni
Team Member 2	Akili Venkata Siva Sai Srikar
Project Name	Machine Learning–Based Credit Card Approval Prediction System
Maximum Marks	4 Marks

Proposed Solution

Introduction

The proposed solution is a Machine Learning–Based Credit Card Approval Prediction System that automates the evaluation of credit card applications using historical financial data and predictive analytics. The system minimizes manual intervention by accurately classifying applicants as eligible or ineligible for credit card approval. A web-based application built using Flask provides an intuitive interface for applicants and administrators, while Machine Learning models deliver fast, consistent, and reliable predictions.

Solution Overview

Applicant

↓

User Registration and Login

↓

Enter Personal and Financial Information

↓

Data Validation and Preprocessing

↓

Machine Learning Prediction Engine
(Logistic Regression, Decision Tree, Random Forest, XGBoost)

↓

Prediction Result Generation

↓

Approval or Rejection Display

↓

Prediction History Stored in Database

↓

Administrator Monitoring and Reporting

Key Features of the Proposed Solution

	Feature	Description
1.	Secure User Authentication	Allows applicants and administrators to register and log in securely using authenticated credentials.
2.	Applicant Data Collection	Collects personal, employment, financial, and credit-related information through an intuitive web interface.
3.	Data Preprocessing	Handles missing values, encodes categorical variables, validates user inputs, and prepares data for Machine Learning prediction.
4.	Machine Learning Prediction	Uses trained classification models such as Logistic Regression, Decision Tree, Random Forest, and XGBoost to predict credit card approval.
5.	Prediction Result	Displays the approval or rejection result instantly along with the predicted eligibility status.
6.	Prediction History	Stores applicant details and prediction outcomes securely for future reference and administrative analysis.
7.	Administrative Dashboard	Allows administrators to review prediction records, monitor application activity, and generate reports.

Advantages of the Proposed Solution

- Automates credit card approval prediction.
- Reduces manual processing time.
- Improves prediction accuracy.
- Supports secure applicant data management.
- Ensures consistent decision-making.
- Enhances customer experience.
- Provides scalable web-based deployment.
- Enables future Machine Learning model improvements.

Conclusion

The proposed Machine Learning–Based Credit Card Approval Prediction System integrates modern web technologies with advanced Machine Learning algorithms to automate the credit approval process. The solution improves decision accuracy, reduces operational workload, enhances customer satisfaction, strengthens data security, and provides a scalable platform suitable for deployment in real-world banking environments.